

Encoder baseline progression — what each addition contributes

new / changed in this step

carried over from previous step

Step 1 — Base baseline

Frozen Whisper-small encoder. Take the last hidden state, mean-pool over time, run a 1-layer FC head.

F1 0.862 BA 0.795

AUROC 0.870

n=635, BA-tuned



Step 2 — + Attention pooling

Replace mean with a learned attention head that scores each frame and computes a softmax-weighted sum — lets the model focus on child-vocal frames.

F1 0.880 BA 0.779

AUROC 0.848

n=635, BA-tuned



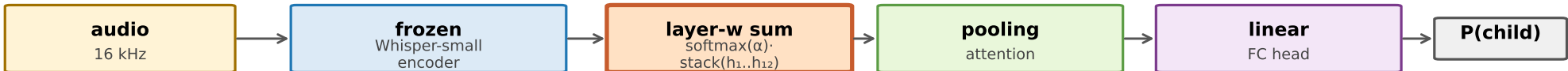
Step 3 — + Layer-weighted sum

Mix all 12 transformer layers with a learned softmax(α) instead of using only the last hidden state — earlier layers carry phonetic info, later layers carry semantics.

F1 0.876 BA 0.776

AUROC 0.886

n=441, BA-tuned



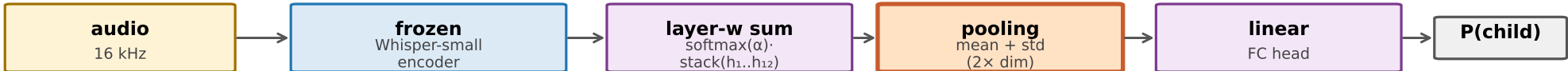
Step 4 — + Statistical pooling (mean + std)

Swap attention for mean+std pooling — the head gets both central tendency and variability across frames (output dim doubles to 1536).

F1 0.850 BA 0.722

AUROC 0.871

n=441, BA-tuned



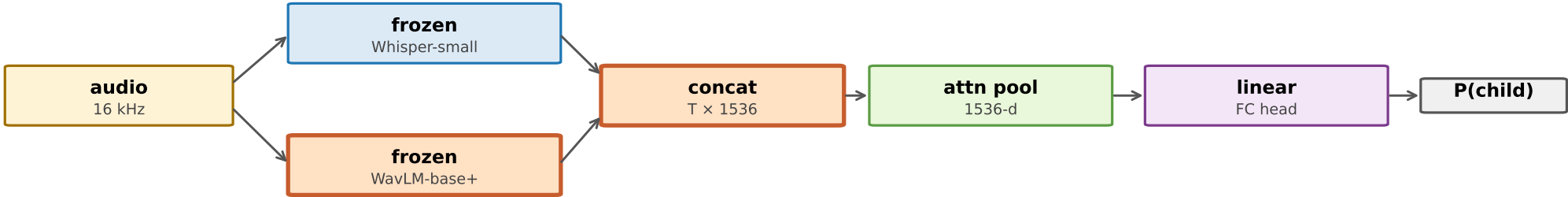
Step 5 — + Fused encoder (Whisper + WavLM)

Add a parallel WavLM-base+ branch; concatenate per-frame embeddings ($T \times 1536$) before pooling — combines Whisper's ASR-style features with WavLM's speaker-style features.

F1 0.870 BA 0.749

AUROC 0.886

n=441, BA-tuned



Step 6 — + Unfreeze last 2 layers of each encoder

Stop freezing the top 2 transformer layers in both branches; train them at $lr=1e-5$ (head still at $lr=1e-3$). Small targeted adaptation, biggest single jump in BA + AUROC.

F1 0.892 BA 0.797

AUROC 0.903

n=441, BA-tuned

